Legend
- Line 66
- Line 67
- Line 68
- Line 69
- Line 70
- Line 71
- Line 72
- Line 73
- Line 74
- Line 75
- Line 76
- Line 77
- Line 78
- Line 79
- Line 80

101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr

4

4" -FL-101-1204-A11A

3" -FL-101-1205-A11A

4"x3"

3"-WP-101-4811-B19A-IH

FROM 101-E-130A/B
WASH WATER
101-1148

48"-P-101-2701-A19A-IH

FROM 101-C-101
CR COL OVHD VAP
101-1127

TW 1212 | PI 1205 | TE 1223 | TT 1223 | FF | TI 1223

NOTE-3,5

NO POCKETS

MAKE CONNECTION AT MID POINT OF HEADER FROM TOP

MIN.

36"-P-101-1210-A19A-IH

NOTE-3,5

36"x40" 48"x40" 40"x36" 48"x40"

NOTE-3,5

36"-P-101-1211-A19A-IH

MIN.

TI 1205 | TT 1205 | TE 1205

PI 1201 | OP

TW 1210

24"-P-101-1209-B1A-IT
TO 101-E-103A/B & 105A/B
CRUDE
101-1113

14"x20"

20"x24"

NOTE-6

NC

FI 1201 A | FI 1201 B | FT 1201 | FF | FE-1201 | 26"-P-101-1212-A19A-IH | FACING | SLOPE 1:500

TI 1201 | TT 1201 | TE 1201 | 12"-P-101-1205-B1A-IT

1.5E2 36.7 | PSV 1201 | 900# 2"x1-1/2" | 2"-SV-101-1201-B1A/T | LO FB | 3/4"

2"-WP-101-1201-B19A-IH

4 | SP 036 | NOTE-4 | B19A A19A | 3/4" | MIN.

2"-P-101-1220-B1A-IT

A11A A19A IH | 3/4"

FI 1202 A | FI 1202 B | FT 1202 | FF | FE-1202 | FACING

TI 1202 | TT 1202 | TE 1202 | 12"-P-101-1206-B1A-IT

1.5E2 36.7 | PSV 1202 | 900# 2"x1-1/2" | 2"-SV-101-1202-B1A/T | LO FB | 3/4"

26"-P-101-1215-A19A-IH | SLOPE 1:500

SP 110 | NOTE-4 | A19A B19A

2"-WP-101-1202-B19A-IH

3"-FL-101-1202-A11A

2" | CP 1210 | CC | SP 010

2"-P-101-1221-B1A-IT

A19A A11A IH

FI 1203 A | FI 1203 B | FT 1203 | FF | FE-1203 | FACING

TI 1203 | TT 1203 | TE 1203 | 12"-P-101-1207-B1A-IT

1.5E2 36.7 | PSV 1203 | 900# 2"x1-1/2" | 2"-SV-101-1203-B1A/T | LO FB | 3/4"

2"-WP-101-1203-B19A-IH

SP 111 | NOTE-4 | A19A B19A

2"-P-101-1222-B1A-IT

3"-FL-101-1203-A11A

FI 1204 A | FI 1204 B | FT 1204 | FF | FE-1204 | FACING

TI 1204 | TT 1204 | TE 1204 | 12"-P-101-1208-B1A-IT

1.5E2 36.7 | PSV 1204 | 900# 2"x1-1/2" | 2"-SV-101-1204-B1A/T | LO FB | 3/4"

26"-P-101-1215-A19A-IH | SLOPE 1:500

SP 112 | NOTE-4 | A19A B19A

2"-WP-101-1204-B19A-IH

3"-FL-101-1204-A11A

2" | CP 1212 | CC | SP 016

NOTE-8

2"-P-101-1223-B1A-IT

A19A A11A IH

MIN.

26"x24"

300#

T1 | T2 | S1 | S2

A | B | C | D

101-E-102A/B/C/D

14"-P-101-1209-B1A-IT

12"-P-101-1201-B1A-IT | 12"-P-101-1202-B1A-IT | 12"-P-101-1203-B1A-IT | 12"-P-101-1204-B1A-IT

TT 1206 | TE 1206 | TI 1206 | PI 1209 | NOTE-8 (TYP.) | SP 008 | CC | 2" | CP 1205

TE 1207 | TT 1207 | TI 1207 | PI 1206 | SP 012 | CC | 2" | CP 1206

TI 1208 | TT 1208 | TE 1208 | PI 1212 | SP 014 | CC | 2" | CP 1207

TE 1209 | TT 1209 | TI 1209 | PI 1215 | SP 018 | CC | 2" | CP 1208

FROM 101-E-101A/B
CRUDE
101-1111

24"-P-101-1113-B1A-IT

NOTE-6

16"x12"

24"x16"

TWO PHASE FLOW — NOTE-5,10 — NO POCKETS

TO 101-A-101 D
24"-P-101-1219-A19A-IS
CR COL OVHD
101-1128

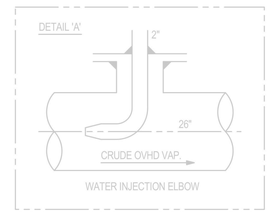
TO 101-A-101 C
24"-P-101-1218-A19A-IS
CR COL OVHD
101-1128

TO 101-A-101 B
24"-P-101-1217-A19A-IS
CR COL OVHD
101-1128

TO 101-A-101 A
24"-P-101-1216-A19A-IS
CR COL OVHD
101-1128

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH 'FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED.
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.



THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

No.	REVISIONS	DATE	DRN	BY	CKD	APPD
4	ISSUED FOR CONSTRUCTION	10.03.22	SC	MW	YD	PV
3	ISSUED FOR CONSTRUCTION	30.09.21	SC	MW	YD	PV
2	ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION	02.09.20	RMB	KG	RS	SG
1	ISSUED FOR ENGINEERING/DESIGN	11.08.20	JO	KG	RS	SG
0	ISSUED FOR REVIEW	13.03.20	LC	KG	RS	SG

DETAILED ENGINEERING CONTRACTOR: FLUOR — FLUOR DANIEL INTERNATIONAL (MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA PROJECTS LIMITED, MUMBAI

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
1	14.12.18	REVISED AS MARKED	PP	UA/AT	GN	AM
0	15.11.18	ISSUED FOR PACKAGE	S.K.L	UA/AT	GN	AM

PMC: ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

OWNER: एच पी सी एल राजस्थान रिफाइनरी लिमिटेड — HPCL RAJASTHAN REFINERY LTD — RAJASTHAN REFINERY PROJECT

पाइपिंग एण्ड इंस्ट्रूमेन्टेशन डायग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

SCALE	JOB NO.	DEPT.	SECTN.	UNIT	DWG. No.	REV.
	B068	79	41	101	1112	1

TPL DRAWING NO. B224-CDU-TPL-101-1112 — REV 4

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DOCUMENT REVIEW DETAILS
Status: V
Comments: Doc. B224-CDU-TPL-101-1112 Rev. 4
1 Approved with manufacture / fabrication as per the commented document. Revised document required.
2 Approved document does not conforms to basic requirements as marked. Resubmit for review.
3
R Received for Records. V Void.
Signature with Date: 24-AUG-2022
Name of Reviewer: UMER AFZAL (C889)